

Algorithm design

→ Teams Classroom

Roberto Grossi

roberto.grossi@unipi.it

agenda.unipi.it

- ① We use randomization as a tool
 - ② We attack problems → hardness
 - ③ Approximation algorithms
 - ④ Parameterized algorithms
 - ⑤ Algorithmic Game Theory → Dr. Filippo Gencì
next two weeks
- } graphs

Temporary: 4-13

Mon March 4 → March 7 { 16:18

March 11 → March 14

Randomization

random(a, b) ($a \leq b$)

↳ $x \in [a..b]$ uniformly with prob. $\frac{1}{b-a+1}$

Building block → concentration bounds

w.h.p. → $pr = \frac{1}{nc}$ where n is the input size

Indicator variable

$$X = \begin{cases} 1 & \text{event occurs with prob.} = p \\ 0 & \text{otherwise} \end{cases}$$

$$E[X + X'] = E[X] + E[X'] \quad \text{even if } X \text{ and } X' \text{ are not independent}$$

$$E[X] = 1 \cdot p + 0 \cdot (1-p) = p = \Pr[X=1]$$

BIRTHDAY PARADOX

$n = 365$ days in a non-leap year

hp each person has been born in a day with uniform prob. $\frac{1}{n}$

$$X_{ij} = \begin{cases} 1 & \text{if both persons } i \text{ and } j \text{ have the same birthday} \\ 0 & \text{otherwise} \end{cases}$$

$$\Pr[X_{ij} = 1] = n \cdot \underbrace{\left(\frac{1}{n} \cdot \frac{1}{n}\right)}_{\substack{\text{same} \\ \text{person} \\ \text{day}}} = \frac{1}{n} = E[X_{ij}]$$

$m = \# \text{ people}$

$$\sum_{1 \leq i < j \leq m} X_{ij} = \# \text{ pairs among } m \text{ people} \\ \text{with the same birthday}$$

$$E \left[\sum_{1 \leq i < j \leq m} X_{ij} \right] = \sum_{1 \leq i < j \leq m} E[X_{ij}] = \sum_{1 \leq i < j \leq m} \frac{1}{n} = \binom{m}{2} \frac{1}{n}$$

↙ expected # of pairs of people with same birthday

$$\binom{m}{2} \frac{1}{n} \geq 1 \Rightarrow m \geq \sim \sqrt{n}$$

$$n = 365 \text{ days} \Rightarrow \sim \sqrt{n} = 27$$